

Full Conditional Distributions

Let $\mathbf{z}$ denote the vector of input estimates, $\mathbf{D} = \text{diag}(d_1, \dots, d_m)$ their sampling covariance matrix, and $\mathbf{X}$ the $m \times j$ covariate matrix. We write $data = \{\mathbf{z}, \mathbf{X}, \mathbf{D}\}$ and use $\dots$ to denote the data and all remaining parameters in a full conditional. The distribution $IG(a, b)$ has density proportional to $x^{-(a+1)} \exp(-b/x)$.

Fay–Herriot model

The model in `fh.R` is

$$\begin{aligned} [\mathbf{z} \mid \boldsymbol{\beta}, \mathbf{v}] &\sim N_m(\mathbf{X}\boldsymbol{\beta} + \mathbf{v}, \mathbf{D}), \\ [\mathbf{v} \mid \sigma^2] &\sim N_m(\mathbf{0}, \sigma^2 \mathbf{I}_m), \end{aligned}$$

with $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1$. The sampler updates $\boldsymbol{\beta}$, $\mathbf{v}$, and σ^2 separately.

Full conditional distribution of $\boldsymbol{\beta}$. The terms involving $\boldsymbol{\beta}$ are

$$\pi(\boldsymbol{\beta} \mid \dots) \propto \exp \left\{ -\frac{1}{2} (\mathbf{z} - \mathbf{X}\boldsymbol{\beta} - \mathbf{v})^\top \mathbf{D}^{-1} (\mathbf{z} - \mathbf{X}\boldsymbol{\beta} - \mathbf{v}) \right\}.$$

Completing the square gives

$$[\boldsymbol{\beta} \mid \mathbf{v}, data] \sim N_j \left(\mathbf{P}_\beta^{-1} \mathbf{X}^\top \mathbf{D}^{-1} (\mathbf{z} - \mathbf{v}), \mathbf{P}_\beta^{-1} \right), \quad \mathbf{P}_\beta = \mathbf{X}^\top \mathbf{D}^{-1} \mathbf{X}.$$

Full conditional distribution of $\mathbf{v}$. Combining the likelihood and random-effect prior gives

$$[\mathbf{v} \mid \boldsymbol{\beta}, \sigma^2, data] \sim N_m \left(\mathbf{P}_v^{-1} \mathbf{D}^{-1} (\mathbf{z} - \mathbf{X}\boldsymbol{\beta}), \mathbf{P}_v^{-1} \right), \quad \mathbf{P}_v = \mathbf{D}^{-1} + \frac{\mathbf{I}_m}{\sigma^2}.$$

Since $\mathbf{P}_v$ is diagonal, the sampler draws the effects independently as

$$[v_i \mid \boldsymbol{\beta}, \sigma^2, data] \sim N \left(\frac{\sigma^2 (z_i - \mathbf{x}_i^\top \boldsymbol{\beta})}{\sigma^2 + d_i}, \frac{\sigma^2 d_i}{\sigma^2 + d_i} \right).$$

Full conditional distribution of σ^2 . Under the flat variance prior,

$$\pi(\sigma^2 \mid \dots) \propto (\sigma^2)^{-m/2} \exp \left\{ -\frac{\mathbf{v}^\top \mathbf{v}}{2\sigma^2} \right\},$$

so

$$[\sigma^2 \mid \mathbf{v}] \sim IG \left(\frac{m}{2} - 1, \frac{\mathbf{v}^\top \mathbf{v}}{2} \right).$$

With the optional $IG(a, b)$ prior, this becomes $IG(m/2 + a, \mathbf{v}^\top \mathbf{v}/2 + b)$.

Taxonomy smoothing

For **ts.R**, let $\mathbf{Z}_r$ be the $m \times n_r$ membership matrix for the n_r nodes at level r . The model is

$$[\mathbf{z} \mid \boldsymbol{\beta}, \mathbf{v}^{(1)}, \dots, \mathbf{v}^{(R)}] \sim N_m \left(\mathbf{X}\boldsymbol{\beta} + \sum_{r=1}^R \mathbf{Z}_r \mathbf{v}^{(r)}, \mathbf{D} \right),$$

$$[\mathbf{v}^{(r)} \mid \sigma_r^2] \sim N_{n_r}(\mathbf{0}, \sigma_r^2 \mathbf{I}_{n_r}).$$

We use $[\boldsymbol{\beta}] \sim N_j(\mathbf{0}, k^2 \mathbf{I}_j)$ and $[\sigma_r^2] \sim IG(a_r, b_r)$. The inputs are standardized before fitting, with defaults $k^2 = 10^4$, $a_r = 3$, and $b_r = 0.6$.

Full conditional distribution of the fixed and random effects. These effects are sampled in a block. Set

$$\boldsymbol{\gamma} = (\boldsymbol{\beta}^\top \quad (\mathbf{v}^{(1)})^\top \quad \dots \quad (\mathbf{v}^{(R)})^\top)^\top, \quad \mathbf{Z} = (\mathbf{X} \quad \mathbf{Z}_1 \quad \dots \quad \mathbf{Z}_R).$$

Writing $q = j + \sum_{r=1}^R n_r$, the prior is $[\boldsymbol{\gamma} \mid \sigma_1^2, \dots, \sigma_R^2] \sim N_q(\mathbf{0}, \boldsymbol{\Lambda}_\gamma^{-1})$, where

$$\boldsymbol{\Lambda}_\gamma = \begin{pmatrix} \mathbf{I}_j/k^2 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{n_1}/\sigma_1^2 & & \mathbf{0} \\ \vdots & & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I}_{n_R}/\sigma_R^2 \end{pmatrix}.$$

Combining the likelihood and prior gives

$$\pi(\boldsymbol{\gamma} \mid \dots) \propto \exp \left\{ -\frac{1}{2}(\mathbf{z} - \mathbf{Z}\boldsymbol{\gamma})^\top \mathbf{D}^{-1}(\mathbf{z} - \mathbf{Z}\boldsymbol{\gamma}) - \frac{1}{2}\boldsymbol{\gamma}^\top \boldsymbol{\Lambda}_\gamma \boldsymbol{\gamma} \right\}.$$

Completing the square yields

$$[\boldsymbol{\gamma} \mid \sigma_1^2, \dots, \sigma_R^2, \text{data}] \sim N_q \left(\mathbf{P}_\gamma^{-1} \mathbf{Z}^\top \mathbf{D}^{-1} \mathbf{z}, \mathbf{P}_\gamma^{-1} \right), \quad \mathbf{P}_\gamma = \mathbf{Z}^\top \mathbf{D}^{-1} \mathbf{Z} + \boldsymbol{\Lambda}_\gamma.$$

Full conditional distributions of $\sigma_1^2, \dots, \sigma_R^2$. For each level,

$$[\sigma_r^2 \mid \mathbf{v}^{(r)}] \sim IG \left(\frac{n_r}{2} + a_r, \frac{(\mathbf{v}^{(r)})^\top \mathbf{v}^{(r)}}{2} + b_r \right).$$

The code draws the reciprocal precision from the equivalent gamma distribution.

Connection to the implementation

The arguments **y** and **d.var** contain $\mathbf{z}$ and the diagonal of $\mathbf{D}$. For a domain without a usable input, its likelihood term is omitted. Equivalently, the matrix calculations use zero precision and a finite placeholder for that domain. Its random effect is still sampled and remains in the variance update.

The Cholesky factor $\mathbf{U}$ satisfies $\mathbf{U}^\top \mathbf{U} = \mathbf{P}_\beta$ in **fh.R** and $\mathbf{U}^\top \mathbf{U} = \mathbf{P}_\gamma$ in **ts.R**. Both samplers use triangular solves to draw from the normal conditionals without forming a matrix inverse. The taxonomy wrapper transforms posterior draws back to the input scale.